

STATS 510: Probability and Distribution Theory

Fall 2025, Prof. Long Nguyen

Text: *Statistical Inference* by George Casella and Roger Berger. Second Edition, Brooks/Cole Cengage Learning.

Course Description: The course will cover essential concepts of probability and distribution theory that are important for statistical inference. The material will be based on the first five chapters from the text including: probability spaces, independence and conditional probability, random variables, joint and conditional distributions, conditional expectation, modeling dependence, transformations, quantiles, order statistics, modes of convergence for sequences of random variables, basic inequalities, laws of small and large numbers, the Central Limit Theorem and the multivariate Normal distribution.

Instructor: Long Nguyen (xuanlong@umich.edu), 243 West Hall.

Office Hours: TBA.

Lectures: Tuesday and Thursday, 11:30am-1pm, CHEM 1200.

GSIs: Sushu Zhang (shushuz@umich.edu) and Jiaxun Li (jasonli@umich.edu).
GSI's office hours: TBA.

Homework and grading policy: Homework is assigned on Canvas every two weeks (about 6 or 7 assignments in total). No late homework is accepted. The weakest homework grade will be dropped from the homework score. There will be two in-class midterm exams and a final exam.

The final grade is based on homework (30%), midterm (30%), and final (40%).

Important Dates:

- *Midterm Exam 1:* September 18 (Thursday) 11:30am–1pm (in class).
- *Midterm Exam 2:* October 21 (Tuesday) 11:30am–1pm (in class).
- *No classes on:* October 14 (Tuesday, Study break) and November 27 (Thursday, Thanksgiving break).
- *Last Lecture:* December 4 (Thursday), 2025.
- *Final Exam:* December 16 (Tuesday) in our usual classroom (CHEM 1200) from 4:00 pm to 6:00 pm, per the official *exam schedule* on: <http://www.umich.edu/~regoff/>. Note that no change can be made to the final exam's date.

Special Accommodations: Should you need special exam accommodations, please schedule your exam through the wonderful Testing Accommodation Center <https://ssd.umich.edu/tac>. If you are registered, you can directly go to:

<https://umich-accommodate.symplicity.com/>

Please write to me (xuanlong@umich.edu) or talk to me in class about your accommodation needs whatever they may be.

Attendance: No formal attendance will be taken. However, the students are expected to come to all lectures and turn in homework regularly. The lectures will not be recorded. Coming to class and actively/humanly engaging in lectures by asking relevant questions will be very helpful to learning the material. As we go along, ask the question "why". From my personal experience it takes many iterations to learn probability theory properly; the deepened understanding will likely be worth the effort.